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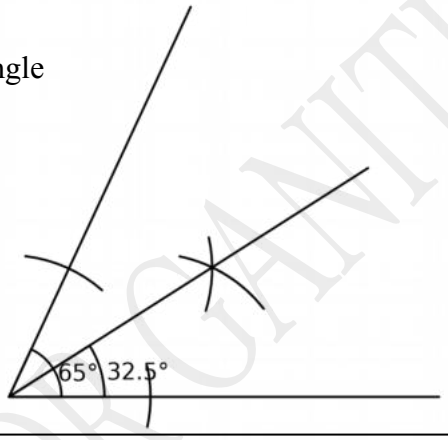
A

SUMMATIVE ASSESSMENT- II 2025 – 26

Class: VIII

MATHEMETICS – ANSWER KEY

E-803

Qn no	Key	Score
	SECTION A	
1	B. 2 [5 times a number $= 6 + 4 = 10$]	1
2	D. $5 - 7$	1
3	D .16	1
4	C. Both statements are true , Statement 2 is the reason of Statement 1	1
	SECTION B	
5	(i) Price of an umbrella $= \frac{900}{3} = 300\ Rs$ (ii) Price of 2 bags $= 2300 - 900 = 1400\ Rs$ Price of a bag $= \frac{1400}{2} = 700\ Rs$	1 1
10	For drawing 65^0 angle For drawing the bisector of this angle 	1 1
7	$\frac{7}{25} = \frac{7 \times 4}{25 \times 4}$ $= \frac{28}{100} = 0.28$ OR $\frac{7}{25} = \frac{7}{5^2} = \frac{7 \times 2^2}{5^2 \times 2^2}$ $= \frac{28}{10^2} = \frac{28}{100} = 0.28$	1 1
8	Number of correctly answered questions $= 5$ Number of wrongly answered questions $= 15$ Total marks $= 5 \times 2 - (15 \times 1) = 10 - 15 = -5$	 1 1
	SECTION C	
9	$AB = 15 \times \frac{10}{5} = 30\ cm$ (i) $AP = 30 \times \frac{2}{10} = 6\ cm$ (ii) $PQ = 30 \times \frac{3}{10} = 9\ cm$ (iii) $AP = 2 \times 6 = 12\ cm$ OR	 1 1 1

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	Take $AP = 2x$, $PQ = 3x$, $QB = 5x$, then $5x = 15 \implies x = \frac{15}{5} = 3$ (i) $AP = 2 \times 3 = 6\ cm$ (ii) $PQ = 3 \times 3 = 9\ cm$ (iii) $AP = 2 \times 6 = 12\ cm$	
10A	In the figure O is the centre of the circle and AB and CD are two chords .Draw a line OP perpendicular to AB and a line OQ perpendicular to CD OP = OQ [Given] Also AP = BP , CQ = DQ [Perpendicular from the centre of a circle to the chord bisects the chord] OA = OC [Radii of a circle are equal] In right triangle OPA , $AP^2 = OA^2 - OP^2$ In right triangle OQC , $CQ^2 = OC^2 - OQ^2$ $= OA^2 - OP^2 = AP^2$ $\therefore AP = CQ \implies 2 \times AP = 2 \times CQ \implies AB = CD$	1 1 1
10B	In the figure O is the centre of the circle and AB is a chord . Draw a line OP perpendicular to AB . $\therefore \angle OPA = \angle OPB = 90^0$ OA = OB [Radii of a circle are equal] In right triangle OPA , $AP^2 = OA^2 - OP^2$ In right triangle OPB , $BP^2 = OB^2 - OP^2$ $= OA^2 - OP^2 = AP^2$ $\therefore AP = BP$	1 1 1
11	Draw a rectangle of sides 7 cm and 5.5 cm For joining the midpoints of the opposite sides For completing the rhombus	1 1 1
12A	(i) $\angle C = 30^0$ [$AB = AC$] (ii) $BD = \frac{7}{2} = 3.5\ cm$. (iii) $\angle BAD = 180 - (90 + 30) = 180 - 120 = 60^0$	1 1 1

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12B

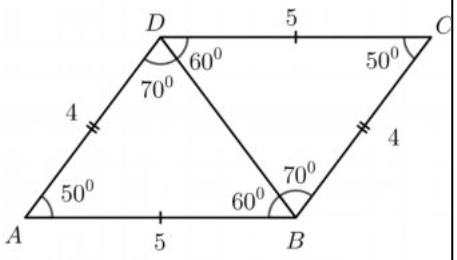
i) Angles of the triangles ABD and BCD are equal.

Because they are equal triangles. 1

[AB = CD , AD = BC , BD = BD]

ii) $\angle A = 50^0$, $\angle ABD = 60^0$, $\angle ADB = 70^0$

[Angles opposite equal sides of equal triangles are equal] 2



SECTION D

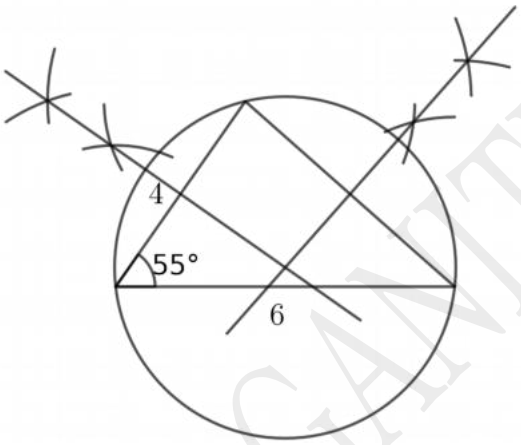
13

For drawing the triangle in the given measurements 2

For drawing the perpendicular bisectors of the sides

For drawing the circle with radius as the distance between a vertex and the point of intersection of the perpendicular bisectors of the sides 1

1



14A

(i)

	Age now	Age after one year
Ammu	x	$x + 1$
Appu	$x + 5$	$x + 6$

1

(ii) $x + 6 = 2(x + 1)$

$x + 6 = 2x + 2 \implies x = 4$ 1

Age of Ammu now = 4

Age of Appu now = 4 + 5 = 9 1

14B

(i) length + breadth $= \frac{60}{2} = 30$ *cm* 1

Take , breadth $= x$ *cm* , then length $= 2x + 3$ *cm*

(ii) $2x + 3 + x = 30 \implies 3x + 3 = 30$ 1

$3x = 27 \implies x = \frac{27}{3} = 9$

$\therefore$ breadth = 9 cm , length $= (2 \times 9) + 3 = 21$ *cm* 1

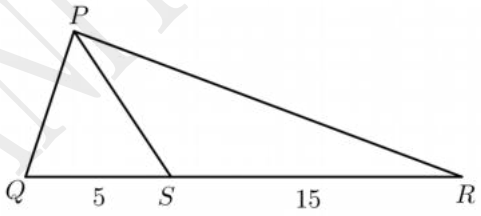
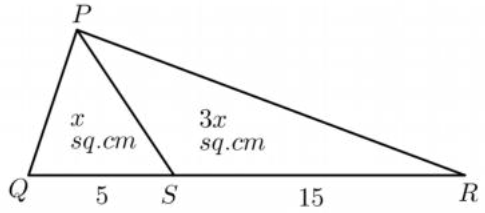
15A

Blue $= 60 \times \frac{2}{5} = 24$ *litres*

Yellow $= 60 \times \frac{3}{5} = 36$ *litres*

Blue is $\frac{2}{3}$ of Yellow. 1

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	To make this ratio 3 : 4 , Blue should be $\frac{3}{4}$ of Yellow $\frac{2}{3}$ $\frac{3}{4}$ $\frac{2\times4}{3\times4}$ $\frac{3\times3}{4\times3}$ Since $\frac{8}{12}$ < $\frac{9}{12}$, we have $\frac{2}{3} < \frac{3}{4}$. So we have to increase blue. If increment in Blue is taken as y *litres* , then $24 + y = \frac{3}{4}\times 36$ $\left[\ \frac{24+y}{36} = \frac{3}{4}\ \right]$ $24 + y = 27 \quad ==> \quad y = 3$ *litres* OR [It can be taken like this too] Blue $= 2x$, Yellow $= 3x$ $2x + 3x = 60 \quad ==> \quad 5x = 60 \quad ==> \quad x = \frac{60}{5} = 12$ $\therefore$ Blue $= 2\times12 = 24$ *litres* , Yellow $= 3\times12 = 36$ *litres*	1 1 1 1
15B	(i) $QS : SR = 5 : 15 = 1 : 3$ (ii) Area of triangle PQS : Area of triangle PSR $= QS : SR = 1 : 3$ Area of triangle PQR $= 4\times10 = 40$ *sq.cm*  [Area of the triangle PQS is $\frac{1}{4}$ of the area of the triangle PQR] OR (ii) Area of triangle PQS : Area of triangle PSR $= QS : SR = 1 : 3$ Area of triangle PQS $= x$ *sq.cm* Area of triangle PSR $= 3x$ *sq.cm* $x = 10$ *sq.cm* Area of triangle PQR $= 4x = 4\times10 = 40$ *sq.cm*  [Area of triangle PQS = 10sq.cm]	1 1 2
16	(i) Sum of outer angles $= 360^0$ (ii) An outer angle $= \frac{360}{10} = 36^0$ [An outer angle $= \frac{360}{\text{Number of the sides}}$] An inner angle $= 180 - 36 = 144^0$ [At a vertex , An inner angle + An outer angle $= 180^0$] (iii) Rectangle OR Square	1 1 1 1

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